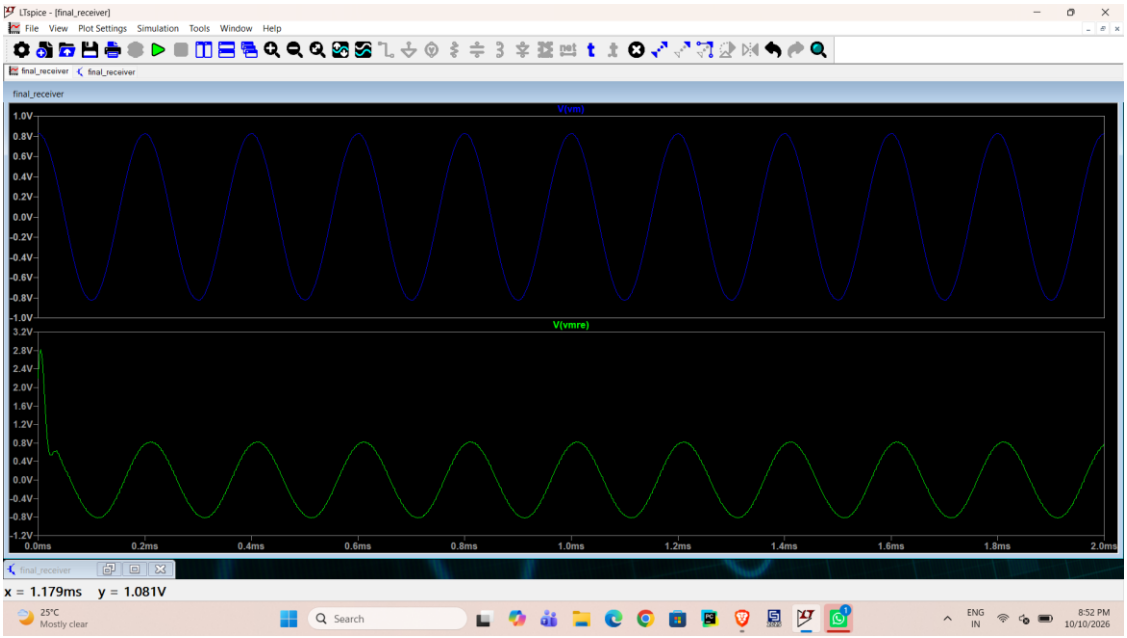
Low-IF AM Receiver: Transistor-Level Analog Baseband in LTspice

A low-IF receiver simulated in LTspice that recovers a 5 kHz message from a 100 kHz AM carrier. The RF source, local oscillators and mixers are ideal behavioral sources. The baseband filters are built from a **two-stage CMOS op-amp designed at transistor level** on a single 3.3 V supply.

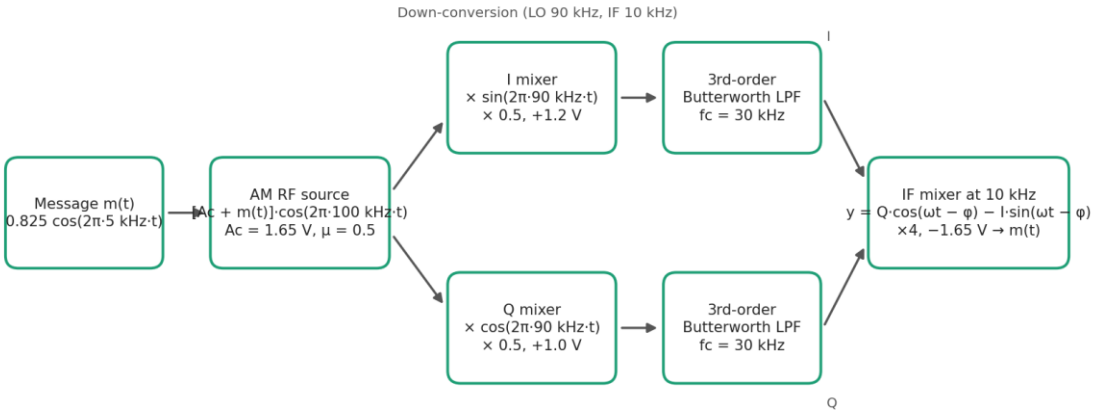


Message versus recovered signal

Top: original message (±0.82 V, 5 kHz). Bottom: message recovered by the receiver (±0.83 V, 5 kHz, about 10 µs delay).

How it works

1. **AM RF signal:** $s(t) = [1.65 + 0.825 \cos(2\pi \cdot 5k \cdot t)] \cos(2\pi \cdot 100k \cdot t)$, modulation index 0.5.
2. **I/Q down-conversion** with a 90 kHz quadrature LO gives a 10 kHz IF (difference) and a 190 kHz sum product.
3. **3rd-order Butterworth active-RC low-pass filters** (30 kHz, one for I and one for Q) remove the sum product.
4. **Second quadrature mixer at the 10 kHz IF**, with a 0.68 rad phase offset that compensates the filter delay, removes the IF carrier. Gain 4 and a 1.65 V subtraction then give back the message.



System block diagram

Key specifications

Parameter	Value
Supply	3.3 V single supply
Carrier / message	100 kHz / 5 kHz, AM with carrier, μ = 0.5
LO / IF	90 kHz / 10 kHz
Low-pass filter	3rd-order Butterworth, active-RC, f_c = 30 kHz
Op-amp	2-stage Miller CMOS, about 80 dB DC gain, 405 μA
Recovered message	about ±0.83 V, 5 kHz

Documentation

File	Contents
`01_project_overview.docx`	Main idea, message and modulation equations, reconstruction maths, specifications, limitations
`02_stage_results.docx`	Simulation result of every stage with a short explanation
`03_circuits_and_design.docx`	Schematics, component values, why they were chosen, role of each component

Repository layout

```
.
├── README.md
├── docs/
│   ├── 01_project_overview.md
│   ├── 02_stage_results.md
│   └── 03_circuits_and_design.md
├── images/                 # LTspice schematics and plots used in the docs
└── ltspice/                # add your .asc / .asy files here (see below)
```

Running the simulation

1. Copy your LTspice files into `ltspice/` and keep them in the **same folder**: `final_receiver.asc` (top level), `lowpassfilter.asc` and `.asy` (the filter block), and the op-amp sheet and symbol.
2. Open `final_receiver.asc` in LTspice. It already contains `.tran 0 2m 0 50n`. Press Run.
3. Plot `V(vm)` and `V(vmre)` to compare the original and the recovered message. Plot `V(vilpf)` and `V(vqlpf)` to see the filtered I/Q.
4. For the filter response, open `lowpassfilter.asc`, use `.ac dec 100 10 1Meg` and plot `V(out)`.
5. For the op-amp, open its sheet and use `.ac dec 100 10 5000Meg`.

The sheets use the generic Level-1 models CMOSN and CMOSP defined on the schematic, so no extra libraries are needed.

Limitations

- Ideal behavioral RF source, LOs and mixers (no transistor-level mixer).
- The 10 kHz second oscillator is synchronized to the carrier by construction. A real receiver needs carrier recovery.
- An image signal at 80 kHz is moved to 20 kHz rather than removed. A baseband filter would reject it.
- The op-amp Miller capacitor of 0.1 pF gives high bandwidth but a low stand-alone phase margin (details in 03_circuits_and_design.docx).

Tools

LTspice (transient and AC analysis), generic Level-1 CMOS models, behavioral B-sources. Results were cross-checked with ngspice.